

Step 2 Results Note

Synthetic Torn/Missing-Region Mask Generation and Validation

Field	Detail
Problem ID	ML-T2-022
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Stage	Stage 2, Step 2 — Synthetic torn-region mask generation (torn-region track)

1. Purpose

Build and validate a generator for irregular, non-rectangular torn/missing-region masks, per the Stage 1 finding that existing systems (e.g. DocRevive) train on regular synthetic occlusions (stamps, scribbles, rectangles) rather than irregular physical tears. Validation goal: confirm masks damage their intended region while leaving surrounding content intact, before this component is used to generate training/evaluation data for Step 3 (reconstruction).

2. Mask Generation: Design Iterations

Three approaches were tried in sequence, each evaluated visually before proceeding:

- Path-with-circles: a jittered random walk with circles stamped along it. Produced an elongated, jagged shape but with visible gaps between circles, not a solid region.
- Jittered-radius polygon (blob): a single center point with radius varied by angle. Produced a solid, gapless shape but compact/round rather than elongated, a weaker visual match for a physical tear.
- Ribbon tear (adopted): a jittered random walk where each point also carries a random width; perpendicular offsets at each point are stitched into one continuous polygon. Produces an elongated, jagged, and fully solid shape, adopted as the final approach.

Edge tears (originating at a page border) and interior tears (originating away from any border) use the same ribbon technique with different starting constraints: edge tears start with the centerline close enough to a border that part of the shape is guaranteed to extend past it; interior tears start with a fully free starting angle inside the page.

3. Word-Targeted Validation

To verify masks damage exactly their intended region, ribbon-style masks were built around known word bounding boxes (from Tesseract's word-level output on the clean image), rather than random placement. Five cases were tested: a single word, two adjacent words, a mask deliberately crossing a word boundary (covering half of one word and half of the next), a larger four-word span, and an edge tear constrained to intersect the text row directly.

Case	Target	Untouched words preserved
Single word	"everything"	17 / 17
Two words	"believe", "everything"	16 / 16
Word boundary	"two" / "equally" (crossing)	17 / 19
Larger span	4 words ("convenient"—"dispense")	14 / 15
Edge tear	left border, intersecting text row	corrupted "To" → "10"; rest of sentence intact

Single- and double-word masks achieved fully surgical damage (all untouched words correctly preserved in OCR output). The boundary-crossing and larger-span masks showed a small, expected bleed into one adjacent word, consistent with OCR reading based on visual continuity rather than strict word boundaries, not treated as a defect.

4. Bug Found and Fixed

Initial version of the text-constrained edge tear produced zero measurable effect on OCR output (byte-identical to the clean transcription). Diagnosis via mask bounding box and pixel count confirmed the mask was being generated but landing entirely in blank margin above the text row, not on the text itself. Root cause: the walk's starting point was correctly anchored to the text row, but nothing constrained its position on subsequent steps, so angle jitter accumulated drift could carry the path out of the text band over the course of the walk.

Fix: clamped the y-coordinate to a padded text-row band after every step of the walk (not just at the start), and reduced per-step angle jitter to limit drift. Re-validated: the corrected edge tear produced a confirmed, visible corruption ("To" → "10") while leaving the remainder of the sentence intact.

A separate, unrelated padding bug was also found and fixed during this step: an early version of the word-targeted mask function computed padding as a percentage of target width, which caused padding to scale up disproportionately for wider multi-word targets, damaging far more surrounding text than intended. This was corrected to a small fixed pixel padding, after which single- and double-word cases reached fully surgical precision (17/17, 16/16 preserved).

5. Interpretation

The mask generator is now validated to do what it claims: it damages a targeted region with high precision for small spans, produces a small and explainable bleed for larger/boundary-crossing spans, and correctly corrupts real text when placed at a page edge. This gives confidence that masks generated for the full torn-region dataset (Step 2 batch generation) will produce a meaningful and controlled testbed for the reconstruction and uncertainty-tagging work planned in Steps 3–4, rather than an unreliable or misleading one.

6. Next Steps

Wrap the validated ribbon-tear generator into a batch pipeline producing multiple masked samples with full metadata (source image, seed, mask path, bounding box, area fraction, edge-vs-interior, generation settings), matching the reproducibility standard established in Step 1. Proceed to Step 3: MLM-based text reconstruction on the resulting masked images.